Binary Information

Binary Name => good kitty
Language => C/C++
Arch => x86x64
Platform => Unix/Linux

```
$ file crack
crack: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=8c584d707909182cb49dab6ebe51cca2217ab1ed, for GNU/Linux 3.2.0, not stripped
```

Analysis

Static Analysis

Below is the disassembly of the **main()**. I have only pasted the essential part

```
..................................................... (SNIP) .....................................................
  0x00005555555554f0 <+580>:  test   al,al
  0x00005555555554f2 <+582>:  je     0x555555555562 <main+694>
  0x00005555555554f4 <+584>:  lea    rdi,[rip+0xb14]        # 0x55555555600f
  0x00005555555554fb <+591>:  call   0x555555555090 <puts@plt>
  0x0000555555555500 <+596>:  mov    rax,QWORD PTR [rsp+0xa8]
  0x0000555555555508 <+604>:  sub    rax,QWORD PTR fs:0x28
  0x0000555555555511 <+613>:  jne    0x555555555570 <main+708>
  0x0000555555555513 <+615>:  mov    eax,0x0
  0x0000555555555518 <+620>:  add    rsp,0xb8
  0x000055555555551f <+627>:  pop    rbx
  0x0000555555555520 <+628>:  pop    rbp
  0x0000555555555521 <+629>:  ret
  0x0000555555555522 <+630>:  mov    edx,DWORD PTR [rsp+0xc]
  0x0000555555555526 <+634>:  add    edx,0x1
  0x0000555555555529 <+637>:  mov    DWORD PTR [rsp+0xc],edx
  0x000055555555552d <+641>:  mov    edx,DWORD PTR [rsp+0xc]
  0x0000555555555531 <+645>:  movsxd rdx,edx
  0x0000555555555534 <+648>:  cmp    rdx,rax
  0x0000555555555537 <+651>:  jge    0x5555555554d9 <main+557>
  0x0000555555555539 <+653>:  mov    edx,DWORD PTR [rsp+0xc]
  0x000055555555553d <+657>:  cmp    edx,0x7
  0x0000555555555540 <+660>:  ja     0x5555555554d9 <main+557>
  0x0000555555555542 <+662>:  mov    ecx,DWORD PTR [rsp+0xc]
  0x0000555555555546 <+666>:  mov    edx,DWORD PTR [rsp+0xc]
  0x000055555555554a <+670>:  movsxd rcx,ecx
```

```
  0x000055555555554d <+673>:   movsxd rdx,edx
  0x0000555555555550 <+676>:   movzx  ebx,BYTE PTR [rsp+rdx*1+0x10]
  0x0000555555555555 <+681>:   cmp    BYTE PTR [rsp+rcx*1+0x60],bl
  0x0000555555555559 <+685>:   je     0x555555555522 <main+630>
  0x000055555555555b <+687>:   mov    BYTE PTR [rsp+0xb],0x0
  0x0000555555555560 <+692>:   jmp    0x555555555522 <main+630>
  0x0000555555555562 <+694>:   lea    rdi,[rip+0xa9b]        # 0x555555556004
  0x0000555555555569 <+701>:   call   0x555555555090 <puts@plt>
  0x000055555555556e <+706>:   jmp    0x555555555500 <main+596>
  0x0000555555555570 <+708>:   call   0x5555555550b0 <__stack_chk_fail@plt>
End of assembler dump.
gef➤  x/s 0x555555556004
0x555555556004: "bad kitty!"
gef➤  x/s 0x55555555600f
0x55555555600f: "good kitty!"
gef➤
```

Program Workflow

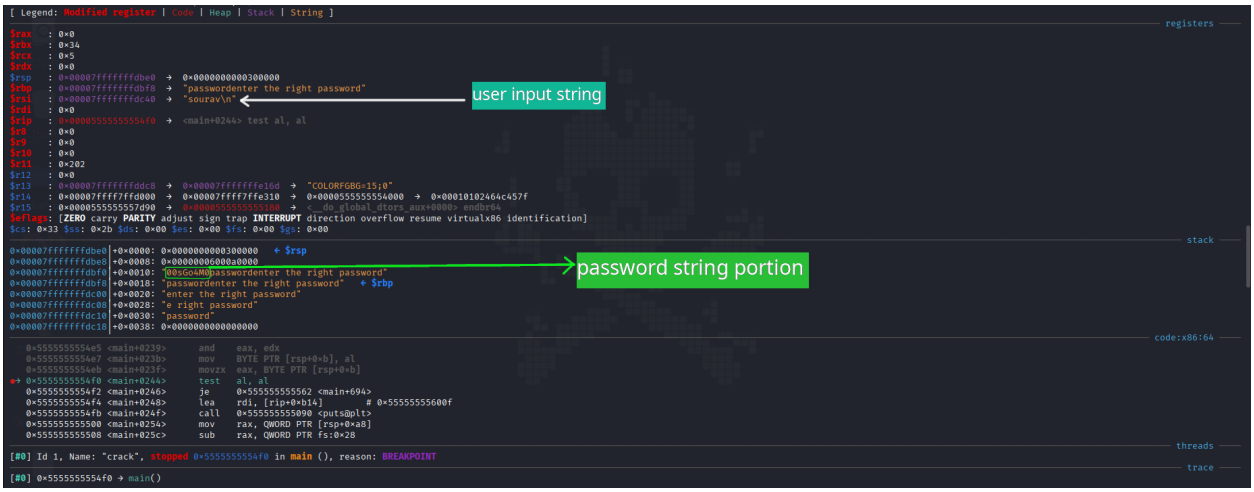
- Takes a password as user input.
- If password is correct **good kitty!** is printed otherwise **bad kitty!** is printed.

Dynamic Analysis

- I setup the breakpoints at **main()** and at 0x00005555555554f0 <+580>: test al,al.

```
gef➤  i b
Num     Type           Disp Enb Address            What
1       breakpoint     keep y   0x00005555555552ac <main>
        breakpoint already hit 1 time
2       breakpoint     keep y   0x00005555555554f0 <main+580>
```

- I ran the program after that, and I noticed somewhere between breakpoint 3 and breakpoint 4 there was a **unique string** which got placed inside the stack. Below is the image of the stack condition when that happens.



Testing our input

```
$ ./crack
enter the right password
00sGo4M0
good kitty!
```

Yeah!!! we were able to solve this crackme challenge.